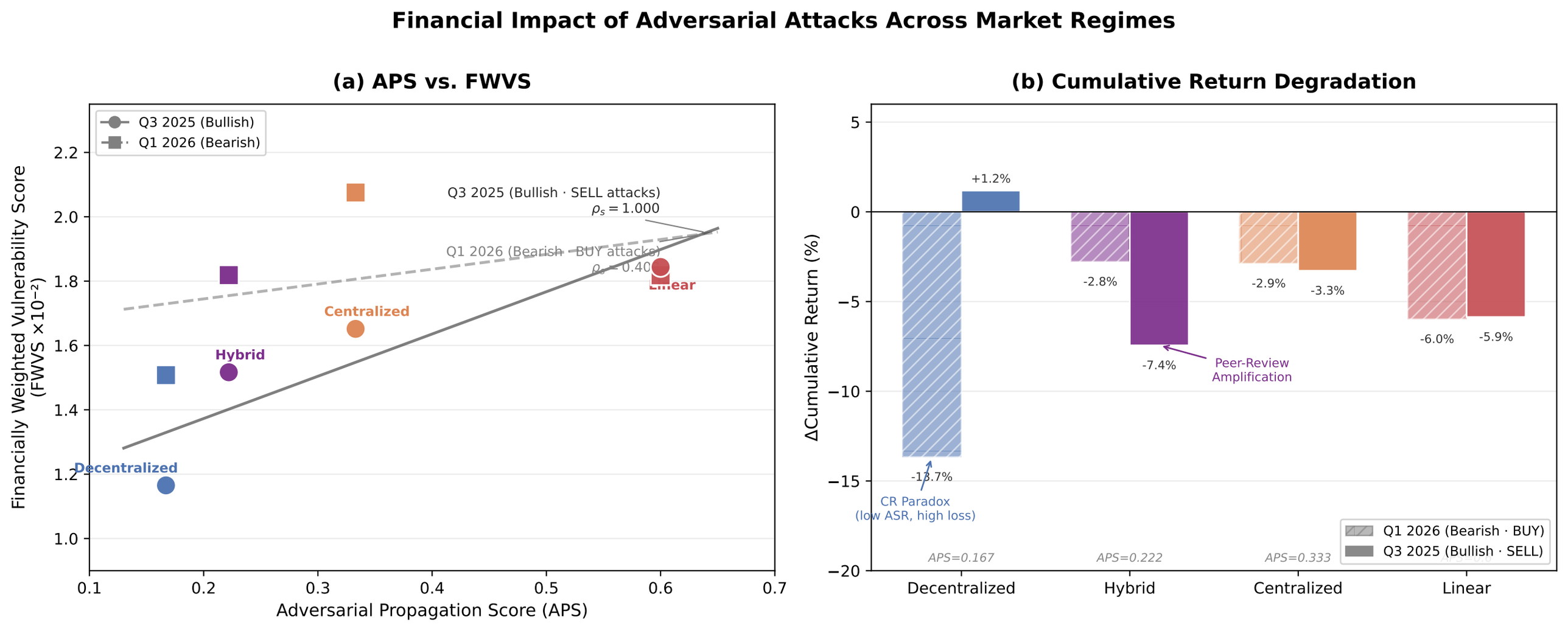
Financial Impact of Adversarial Attacks Across Market Regimes
(a) APS vs. FWVS
Q3 2025 (Bullish)
Q1 2026 (Bearish)
Q3 2025 (Bullish · SELL attacks)
$\rho_s = 1.000$
Q1 2026 (Bearish · BUY attacks)
$\rho_s = 0.400$
Centralized
Hybrid
Decentralized
Linear
Financially Weighted Vulnerability Score (FWVS ×10⁻²)
Adversarial Propagation Score (APS)
2.2
2.0
1.8
1.6
1.4
1.2
1.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
(b) Cumulative Return Degradation
ΔCumulative Return (%)
5
0
−5
−10
−15
−20
+1.2%
-13.7%
CR Paradox
(low ASR, high loss)
-2.8%
-7.4%
Peer-Review
Amplification
-2.9%
-3.3%
-6.0%
-5.9%
APS=0.167
APS=0.222
APS=0.333
Decentralized
Hybrid
Centralized
Linear
Q1 2026 (Bearish · BUY)
Q3 2025 (Bullish · SELL)